Sketchbook Upper Bounds

ABSTRACT

CCS CONCEPTS

• **Theory of computation** → **Bloom filters and hashing**; • **Information systems** → **Unidimensional range search**.

KEYWORDS

Range Filter, Dynamic Data Structure, Data Growth, Scalability.

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1 THEORETICAL ANALYSIS AND RESULTS

We define $\mathsf{PoE}_{\mathsf{CMS}}(d, w, N)$ to denote the extra error we incur in CMS by not knowing the ℓ_1 norm of the stream in advance. Namely, we compare the error of Sketchbook with that of a CMS with d arrays of length w after it has processed a stream of ℓ_1 norm N.

Lemma 1.1. *In the CMS sketch, let $\mathcal{E}_{w^*}$ denote the error we get when we employ w^* as a size function. Then, during epoch i:*

$$\mathsf{E}\left[\mathcal{E}_{w^*}\right] \leq \frac{1}{w_i^*} \cdot \sum_{j=0}^{i} 2^{i-j} \cdot (N_{j+1} - N_j)\,,$$

where N_j denotes the ℓ_1 norm of the stream at the start of epoch j.

Lemma 1.2. *In the CS sketch, let $\mathcal{E}_{w^*}$ denote the error we get when we employ w^* as a size function. Then, during epoch i:*

$$Var\left[\mathcal{E}_{w^*}\right] \leq \frac{1}{w_i^*} \cdot \sum_{j=0}^{i} 2^{i-j} \cdot (F_{j+1} - F_j)\,,$$

where F_j denotes the squared ℓ_2 norm of the stream at the start of epoch j.

this needs more work, because it does involve arguing about the probability of deviating from the expected value..

2 PROOF OF THEOREM FOR INSERTION-ONLY STREAMS

We will do the analysis assuming a general size function w^* and then discuss how the analysis proceeds for specific values of w^*. Note that both CS and CMS share the same general structure: they contain d counter arrays of size w.

Fix some size function w^*. Note that we perform doubling each time the function w^* doubles. In the CMS case, w^* is a function of N, the ℓ_1 norm of the stream. In the CS case, w^* is a function of the ℓ_2 norm of the stream. In either case, we can track the moments *in time* in which the value of the function doubles. Formally, instead of thinking of w^* as a function of the the ℓ_1 or ℓ_2 norms of the stream, we think of it as a function of time such that, at any point in time t, $w^*(t)$ is the number of counters the sketch uses at time t.

Now let T_0 be the first time the sketch is instantiated (i.e., this sketch has d counters and $w^*(T_0)$). We then define T_i for $i \geq 1$ to be the first moment in time when the length of each array doubles, i.e., $w^*(T_i) = 2 \cdot w^*(T_{i-1})$. Note that, in between times T_i and T_{i+1}, the length of each array stays the same (i.e., it is equal to $w^*(T_i)$). We call this period of time epoch i and denote by $\mathcal{S}_i$ the *active* sketch during this epoch (i.e., the one that has arrays of length $w^*(T_i)$). According to our design, updates to the stream during Epoch i are executed directly on $\mathcal{S}_i$. For the purposes of executing queries fast, we also aggregate all previous sketches into the active sketch.

Formally, let (x, Δ) be an update that occured in some epoch $0 \leq j < i$ before the current epoch. In the aggregation step, this update operation is replicated on all the counters in $\mathcal{S}_i$ that x could have hashed into. Namely, if x hashed to an index α in epoch j for one of the arrays, then, during epoch i, all of the 2^{j-1} counters $\alpha \circ \{0,1\}^{j-i}$ ingest the update (x, Δ), where $\circ$ denotes concatenation and $\{0,1\}^{j-i}$ denotes all binary strings of length $j-1$. From the perspective of fixed counter, what this means is that the counter must also ingest previous updates from elements that don't technically hash into it.

Formally, let α be a counter in epoch i, and assume for simplicity that we only have one array in the sketch (i.e., $d = 1$ and $h_1(x) = h(x)$). This counter ingests the following operations:

- updates (x, Δ) from epoch i and for which $h(x) = \alpha$. This event occurs with probability $1/w^*(T_i)$ since there are $w^*(T_i)$ counters in the array. For ease of analysis, we define $\mathcal{J}_{i,\alpha}(x)$ to be the indicator random variable that is 1 if this event occurs, and 0 otherwise. Then $\Pr\left[\mathcal{J}_{i,\alpha}(x) = 1\right] = 1/w^*(T_i)$.
- updates (x, Δ) from epoch $i-1$ that, during epoch $i-1$, hashed into a counter that matches the leftmost $\log w^*(T_i) - 1$ bits of α in the binary representation (meaning, their current $h(x)$ value differs from α in at most the last 1 bit of their binary representation). This event occurs with probability at most $2/w^*(T_i)$. We emphasize that this is merely an upper bound: the additional factor of 2 accounts for the mistakes that the aggregation step makes, i.e., processing updates from previous epoch that would not have hashed into α in the current epoch.
- in general, updates (x, Δ) from epoch $j < i$ and for which $h(x)$ differs from α only in the last $i - j$ bits of its binary representation. This event occurs with probability at most $2^{i-j}/w^*(T_i)$. Similarly as before, we define $\mathcal{J}_{j,\alpha}(x)$ to be the indicator random variable that is 1 if this event occurs, and 0 otherwise. Then $\Pr\left[\mathcal{J}_{j,\alpha}(x) = 1\right] \leq 2^{i-j}/w^*(T_i)$. Similarly as before, note that this is merely an upper bound: for x such

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that $h(x) = \alpha$, the $\Pr\left[\mathcal{J}_{j,\alpha}(x) = 1 | h(x) = \alpha\right] = 1$ no matter when the update to x was processed in the stream.

While updates from previous epochs have a higher chance of hashing into α than updates in epoch i, they are nevertheless bounded in number, i.e., during epoch i, we have $T_{i+1} - T_i$ insertions. In the following, we bound the effect this has on each counter. We first cluster all the insertions that occur during a given epoch into one big insertion per element. Formally, let $f_i(x)$ denote the total count of updates of the form (x, Δ) that occur in the stream during epoch i. Note that $\sum_x f_j(x) = T_{j+1} - T_j$ for all $j \geq 0$.

Bounds for CMS. In the case of CMS, each counter gathers the sum of frequencies of elements that hash into it. Namely, the random variable α can be expressed as:

$$\alpha = \sum_{j=0}^{i} \sum_{x} \mathcal{J}_{j,\alpha}(x) \cdot f_j(x) \, .$$

By linearity of expectation, we get that:

$$E[\alpha] = \sum_{j=0}^{i} \sum_{x} \frac{2^{i-j}}{w^*(T_i)} \cdot f_j(x) = \frac{1}{w^*(T_i)} \cdot \sum_{j=0}^{i} \sum_{x} 2^{i-j} \cdot f_j(x)$$
$$= \frac{1}{w^*(T_i)} \cdot \sum_{j=0}^{i} 2^{i-j} \sum_{x} f_j(x) = \frac{1}{w^*(T_i)} \cdot \sum_{j=0}^{i} 2^{i-j} \cdot (T_{j+1} - T_j) \, .$$

In other words, the increase in error we get because of aggregating the sketches is dominated by the term $\sum_{j=0}^{i} 2^{i-j} \cdot (T_{j+1} - T_j)$. This is where the specific size function w^* we employ has an effect, since it translates into bounds on T_i (and specifically their relative sizes). We now consider various special cases and discuss their respective behavior:

Setting $w^*(N) = N/\epsilon$ in CMS. In this case, we start with the original sketch $\mathcal{S}_0$ with a capacity for T_0 elements. The size function doubles whenever the number of elements doubles. In this case $T_{j+1} = 2 \cdot T_j$ for all $j \geq 0$ and, in general, $T_j = 2^j \cdot T_0$ for all $j \geq 0$. The additional error term then becomes:

$$\sum_{j=0}^{i} 2^{i-j} \cdot (T_{j+1} - T_j) = \sum_{j=0}^{i} 2^{i-j} \cdot (2^{j+1} \cdot T_0 - 2^j \cdot T_0)$$
$$= \sum_{j=0}^{i} 2^{i-j} \cdot 2^j \cdot T_0 = i \cdot 2^i \cdot T_0 \, .$$

Since the ℓ_1 norm of the stream doubles in each epoch, we have that $2^i \cdot T_0 = T_i$ and so, in expectation,

$$E[\alpha] = \frac{1}{w^*(T_i)} \cdot i \cdot T_i = i/\epsilon \, .$$

This is also an upper bound on the expected error we get for an element x. blah blah

Setting $w^*(N) = N^{1-c}/\epsilon$ in CMS, for some $0 < c < 1$. In this case, the size function doubles more rarely, whenever $T_{i+1} = 2^{1/(1-c)} \cdot T_i$. Since $0 < c < 1$, note that this implies that T_{i+1} is bigger than $2T_i$. This difference is why we avoid the extra dependency on i in the error term. Formally, we have that $T_i = 2^{i/(1-c)} \cdot T_0$ and thus:

$$\sum_{j=0}^{i} 2^{i-j} \cdot (T_{j+1} - T_j) = \sum_{j=0}^{i} 2^{i-j} \cdot (2^{(j+1)/(1-c)} \cdot T_0 - 2^{j/(1-c)} \cdot T_0)$$
$$= \sum_{j=0}^{i} 2^{i-j} \cdot 2^{j/(1-c)} \cdot T_0 \cdot (2^{1/(1-c)} - 1)$$
$$= 2^i \cdot T_0 \cdot (2^{1/(1-c)} - 1) \cdot \sum_{j=0}^{i} 2^{j/(1-c)-j} \, .$$

For simplicity, we define $2^{1/(1-c)-1} = a$ and observe that:

$$\sum_{j=0}^{i} 2^{j/(1-c)-j} = \sum_{j=0}^{i} a^j = (a^{i+1} - 1)/(a - 1) \, .$$

We then get that:

$$\sum_{j=0}^{i} 2^{i-j} \cdot (T_{j+1} - T_j) = 2^i \cdot T_0 \cdot (2^{1/(1-c)} - 1) \cdot \sum_{j=0}^{i} 2^{j/(1-c)-j}$$
$$= 2^i \cdot T_0 \cdot (2a - 1) \cdot (a^{i+1} - 1)/(a - 1)$$
$$\leq 2^{i+1} \cdot T_0 \cdot (a^{i+1} - 1) \leq 2^{i+1} \cdot T_0 \cdot a^{i+1} \, .$$

At this point, it is worth mentioning that the expected error we are comparing against is $T_i/w^*(T_i)$, i.e., the error that we get by aggregating is close to the error we would get if we didn't expand and just allocated S_i as a sketch from the beginning. To this end, we first notice that $2a = 2^{1/(1-c)}$ and so

$$2^{i+1} \cdot T_0 \cdot a^{i+1} = 2^{(i+1)/(1-c)} \cdot T_0 = 2^{1/(1-c)} \cdot T_i \, .$$

Therefore, the expected error becomes:

$$E[\alpha] \leq \frac{1}{w^*(T_i)} \cdot \sum_{j=0}^{i} 2^{i-j} \cdot (T_{j+1} - T_j) \leq 2^{1/(1-c)} \cdot \frac{T_i}{w^*(T_i)} \, ,$$

as desired. We note that the above holds whenever $c < 1$. In particular, c does not need to be a constant.

Streams with deletions. to be filled

Bounds for CS. In the case of CS, we are interested in describing the behavior of the random variable $\hat{f}(z) = \alpha \cdot \sigma(z)$ for any query z, where $\sigma(z)$ is the associated sign function (for simplicity, we again assume that $d = 1$). Using the notation we introduced before, the random variable α can be expressed as:

$$\alpha = \sum_{j=0}^{i} \sum_{x} \mathcal{J}_{j,\alpha}(x) \cdot f_j(x) \cdot \sigma(x) = \sum_{x} C(x) \, ,$$

where $C(x) = \sum_{j=0}^{i} \mathcal{J}_{j,\alpha}(x) \cdot f_j(x) \cdot \sigma(x)$ denotes the contribution throughout the epoch of updates to x.

We begin by computing the expected value of the estimator $\hat{f}(z)$ as in the classical CS analysis that shows that in expectation, $\alpha \cdot \sigma(z)$ is an unbiased estimator, i.e. $E[\alpha \cdot \sigma(z)] = f(z)$. By linearity of expectation

$$\mathsf{E}[\hat{f}(z)] = \sum_{x \neq z} \mathsf{E}\left[C(x) \cdot \sigma(z)\right] + \mathsf{E}\left[C(z)\sigma(z)\right] = \sum_{j=0}^{i} f_j(z) = f(z) \, ,$$

where the last step uses the fact that $\mathsf{E}\left[C(x)\cdot\sigma(z)\right]=0$ by construction and that $\Pr\left[\mathcal{J}_{j,\alpha}(z)\right]=1$ since $h(z)=\alpha$. We will now bound the variance of this estimator, and in particular, we will be interested in how much bigger it becomes when compared to $\sqrt{F_i/w^*(T_i)}$, where F_i is the squared ℓ_2 norm of the stream at time T_i. Using a similar argument when expanding the variance of the estimator, we get that:

$$\begin{aligned}\mathsf{Var}\left[\hat{f}(z)\right] &= \mathsf{E}\left[\left(\sum_x C(x)\sigma(z)\right)^2\right]-(f(z))^2\\ &= \sum_x \mathsf{E}\left[(C(x))^2\right]+\sum_x\sum_{y\neq x}\mathsf{E}\left[C(x)C(y)\right]-(f(z))^2\\ &= \sum_x \mathsf{E}\left[(C(x))^2\right]-(f(z))^2\,,\end{aligned}$$

where we use the fact that $\mathsf{E}\left[\sigma(x)\cdot\sigma(y)\right]=0$ for any $x\neq y$ to establish that $\mathsf{E}\left[C(x)C(y)\right]=0$. We now bound $\mathsf{E}\left[(C(x))^2\right]$. Recall that we defined $C(x)=\sum_{j=0}^{i}\mathcal{J}_{j,\alpha}(x)\cdot f_j(x)\cdot\sigma(x)$. Then:

$$\begin{aligned}\mathsf{E}\left[(C(x))^2\right] &= \mathsf{E}\left[\left(\sum_{j=0}^{i}\mathcal{J}_{j,\alpha}(x)\cdot f_j(x)\right)^2\right]\\ &= \sum_{j=0}^{i}\mathsf{E}\left[\left(\mathcal{J}_{j,\alpha}(x)\cdot f_j(x)\right)^2\right]+\\ &+2\cdot\sum_{j=0}^{i-1}\sum_{k=j+1}^{i}\mathsf{E}\left[\mathcal{J}_{j,\alpha}(x)\cdot f_j(x)\cdot\mathcal{J}_{k,\alpha}(x)\cdot f_k(x)\right]\\ &= \sum_{j=0}^{i} f_j(x)^2\cdot\mathsf{E}\left[\mathcal{J}_{j,\alpha}(x)\right]+\\ &+2\cdot\sum_{j=0}^{i-1}\sum_{k=j+1}^{i} f_j(x)\cdot f_k(x)\cdot\mathsf{E}\left[\mathcal{J}_{j,\alpha}(x)\cdot\mathcal{J}_{k,\alpha}(x)\right]\,.\end{aligned}$$

For bounding the $\mathsf{E}\left[\mathcal{J}_{j,\alpha}(x)\right]$, we can use the fact that $\mathsf{E}\left[\mathcal{J}_{j,\alpha}(x)\right]\leq 2^{i-j}/w^*(T_i)$. For $\mathsf{E}\left[\mathcal{J}_{j,\alpha}(x)\cdot\mathcal{J}_{k,\alpha}(x)\right]$, we note that the product of the two indicator random variables is 1 only when both of them are 1. Having said that, the two random variables are not independent: if $\mathcal{J}_{k,\alpha}(x)=1$, then $\mathcal{J}_{j,\alpha}(x)=1$ for any $k\geq j$. Namely, if during epoch k, x hashed into a counter that was a prefix of α, then the same would have happened during epoch $j\leq k$. We get that:

$$\mathsf{E}\left[\mathcal{J}_{j,\alpha}(x)\cdot\mathcal{J}_{k,\alpha}(x)\right]=\mathsf{E}\left[\mathcal{J}_{k,\alpha}(x)\right]\,.$$

The whole sums becomes:

$$\begin{aligned}\sum_{j=0}^{i-1}\sum_{k=j+1}^{i} f_j(x)\cdot f_k(x)\cdot\mathsf{E}\left[\mathcal{J}_{k,\alpha}(x)\right] &= \sum_{k=1}^{i}\sum_{j=0}^{k-1} f_j(x)\cdot f_k(x)\cdot\mathsf{E}\left[\mathcal{J}_{k,\alpha}(x)\right]\\ &= \sum_{k=1}^{i} f_k(x)\cdot\mathsf{E}\left[\mathcal{J}_{k,\alpha}(x)\right]\cdot\sum_{j=0}^{k-1} f_j(x)\\ &= \sum_{k=1}^{i} f_k(x)\cdot\mathsf{E}\left[\mathcal{J}_{k,\alpha}(x)\right]\cdot f_{\leq k-1}(x)\end{aligned}$$

where, for simplicity, we denoted $f_{\leq i}(x)=\sum_{j=0}^{i} f_i(x)$ the sum of frequencies accumulated by x through epochs 0 to i and set $f_{\leq -1}(x)=0$. Then:

$$\begin{aligned}\mathsf{E}\left[(C(x))^2\right] &= \sum_{j=0}^{i}\left(f_j(x)^2+2f_j(x)\cdot f_{\leq j-1}(x)\right)\cdot\mathsf{E}\left[\mathcal{J}_{j,\alpha}(x)\right]\\ &= \sum_{j=0}^{i}\left(f_{\leq j}(x)^2-f_{\leq j-1}(x)^2\right)\cdot\mathsf{E}\left[\mathcal{J}_{j,\alpha}(x)\right]\,.\end{aligned}$$

While the term $f_{\leq j}(x)^2-f_{\leq j-1}(x)^2$ might seem contrited, we note that $\sum_x f_{\leq j}(x)^2-f_{\leq j-1}(x)^2=F_j-F_{j-1}$ denotes the increase in the squared ℓ_2 norm of the stream vector as we move from epoch $j-1$ to epoch j. We thus get that:

$$\begin{aligned}\mathsf{Var}\left[\hat{f}(z)\right] &= \sum_x \mathsf{E}\left[(C(x))^2\right]-(f(z))^2\\ &= \sum_x\sum_{j=0}^{i}\left(f_{\leq j}(x)^2-f_{\leq j-1}(x)^2\right)\cdot\mathsf{E}\left[\mathcal{J}_{j,\alpha}(x)\right]-(f(z))^2\\ &\leq \sum_{j=0}^{i}\left(F_j-F_{j-1}\right)\cdot\frac{2^{i-j}}{w^*(T_i)}\,.\end{aligned}$$

At this point, it becomes perhaps more natural to understand why we defined the size function to depend on the squared ℓ_2 norm of the stream vector. We now discuss different bounds we get on the variance depending on the size function.

Setting $w^*(F)=F/\epsilon$ in CS. In this case, we start with the original sketch $\mathcal{S}_0$ that allows streams for which their squared ℓ_2 norm is at most some F_0. The size function doubles whenever the squared ℓ_2 norm doubles. In this case $F_{j+1}=2\cdot F_j$ for all $j\geq 0$ and, in general, $F_j=2^j\cdot F_0$ for all $j\geq 0$. In this case, we get that:

$$\begin{aligned}\mathsf{Var}\left[\hat{f}(z)\right] &\leq \sum_{j=0}^{i}\left(F_j-F_{j-1}\right)\cdot\frac{2^{i-j}}{w^*(T_i)}\\ &\leq \frac{1}{w^*(T_i)}\cdot\left(F_0\cdot 2^i+\sum_{j=1}^{i}\left(2^j-2^{j-1}\right)\cdot F_0\cdot 2^{i-j}\right)\\ &\leq \frac{1}{w^*(T_i)}\cdot\left(F_0\cdot 2^i+\sum_{j=1}^{i}2^{j-1}\cdot F_0\cdot 2^{i-j}\right)\\ &\leq \frac{1}{w^*(T_i)}\cdot\left(F_0\cdot 2^i+\sum_{j=1}^{i}F_0\cdot 2^{i-1}\right)\\ &\leq \frac{1}{w^*(T_i)}\cdot\left(F_i+(i-1)\cdot F_i/2\right)=\frac{F_i}{w^*(T_i)}\cdot\frac{i+1}{2}\,.\end{aligned}$$

Setting $w^*(F)=F^{1-c}/\epsilon$ in CS, for some $0<c<1$. In this case, the size function doubles more rarely, whenever $F_{i+1}=2^{1/(1-c)}\cdot F_i$. Since $0<c<1$, note that this implies that F_{i+1} is bigger than $2F_i$. This difference is why we avoid the extra dependency on i in the error term. Formally, we have that $F_i=2^{i/(1-c)}\cdot F_0$ and thus:

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$$\sum_{j=0}^{i} 2^{i-j} \cdot (F_{j+1} - F_j) = \sum_{j=0}^{i} 2^{i-j} \cdot (2^{(j+1)/(1-c)} \cdot T_0 - 2^{j/(1-c)} \cdot T_0)$$
$$= \sum_{j=0}^{i} 2^{i-j} \cdot 2^{j/(1-c)} \cdot T_0 \cdot (2^{1/(1-c)} - 1)$$
$$= 2^i \cdot T_0 \cdot (2^{1/(1-c)} - 1) \cdot \sum_{j=0}^{i} 2^{j/(1-c)-j} \ .$$